CS 411: ARTIFICIAL INTELLIGENCE
FALL 2026

COLLEGE OF ENGINEERING, UIC
CS 411: ARTIFICIAL INTELLIGENCE (3 HOURS /4 HOURS)

I. Instructor & Course Details

Instructor Name

Name: Myo Thida
Email address: mthida@uic.edu
Drop-In Hours (in-person): Mon & Wed: 3:30-4:30 PM,
Drop-In Hours location: CDRLC 3458

TA Help Hours

Name: Saran Pandi
Email address: spand43@uic.edu
Drop-In Hours (in-person): Mon & Wed**:** 1:00-2:00 PM (CDRLC 2404)

LMS Course Site:

UIC Course Site: https://canvas.uic.edu/courses/56080

Course Calendar

The course calendar is embedded in LMS (Canvas): Course Calendar Link

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. For all technical questions about Canvas, email the Learning Technology Solutions team at LTS@uic.edu.

Course Modality and Schedule

This course is taught **ON CAMPUS**.

- **Date & Time**: Mon and Wed (05:00 PM - 06:15 PM)
- **Venue**: CDRLC **2406**

II. Course Information

Catalog Course Description and Prerequisite/Corequisite Statement

Definitions of intelligence and AI, autonomous agents, architectures of agents and types of environments, search techniques, adversarial search, search in complex environments, logic and probability, learning, generative AI.

Course Information: 3 hours (Undergraduate) 4 hours (Graduate).

Prerequisite(s): Grade of C or better in CS 251.

Note: Extensive computer use required.

Growth Mindset:
Course materials and assignments can be complex and challenging, but they are crucial to your intellectual and personal growth and development. There are times you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, visit the tutoring center, and contact faculty when struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

Course Goals and Learning Outcomes

By the end of the class, students will be able to:

- Formulate real-world problems as AI search or reasoning problems. (ABET O1)
- Apply AI theories to real-world problems. (ABET O6)
- Design and implement AI algorithms and evaluate their performance. *(ABET O2, & O6)*
- Evaluate the ethical, legal, and societal impacts of AI technologies. (ABET O4)

Required and Recommended Course Materials

Recommended References (Optional):

- **Stuart J. Russell and Peter Norvig,** Artificial Intelligence: A Modern Approach (4th Edition)
    - Online: https://aima.cs.berkeley.edu/
    - E-book: Pearson: ISBN 9780137505135

Required Technology

A personal laptop running Windows 10 or later, macOS 13 or later, or a modern Linux distribution, and a reliable broadband Internet connection are required for this course.

Students will use **GitHub** for version control, assignment submission, and collaboration throughout the semester. A GitHub account is required. Programming assignments and AI projects will primarily be developed in **Python** using tools such as **Jupyter Notebook**, **Visual Studio Code (VS Code)**, or **GitHub Codespaces**, depending on the course activities.

Installation and use of **Visual Studio Code (VS Code)** is strongly encouraged for students who wish to develop more advanced programming, debugging, and project management skills in a professional development environment.

Students who do not have access to the required hardware or Internet connectivity should contact the appropriate university support services to inquire about available laptop loan or connectivity assistance programs.

Respect for Copyright

Please protect the copyright integrity of all course materials and content. Please do not upload course materials not created by you onto third-party websites or share content with anyone not enrolled in our course.

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

Grading Policy and Point Breakdown

- **In-Class Quizzes** – 18%
- **In-Class Labs** – 24%
- **3 In-class and one Final Exams** (4 Exam in Total) – 34%
- **Individual Projects** (3 Projects in Total) – 24%

Category	Weight	Number	Drop Allowed	Make-up
Reading Assignments	Not graded	N/A	N/A	No
In-Class Quizzes	18%	8	Lowest 2 quizzes	No
In-Class Labs	24%	10	Lowest 2 labs	No
Exams	34%	4	No	Yes (1)
Individual Projects	24%	3	No	Yes (1)

Policy for Missed or Late Work

Reading Assignments

- Weekly reading assignments will be assigned to help students prepare for lectures, class discussions, quizzes, and programming activities.
- Unless otherwise specified, reading assignments are due **Sunday at 11:59 PM**.
- Completing the assigned readings is strongly recommended, as the material may be assessed on quizzes, exams, and other course activities.

In-Class Quizzes

- There are **8 Quizzes** in total.
- Quizzes will open on **Canvas** at the beginning of class.
- Students must complete quizzes within the **first 15-20 minutes** of class.
- Late submissions and make-up quizzes are **not accepted**.
- The **lowest 2 quiz scores** will be automatically dropped at the end of the semester.

In-Class Labs

- There are **10 labs** in total. Each lab may be either a **programming lab** or a **conceptual**.
- Labs will be assigned during class.
- All labs must be completed and submitted by **06:30 PM** on the day they are assigned.
- Late submissions and make-up labs are **not accepted**.
- The **lowest 2 lab scores** will be automatically dropped at the end of the semester.

Exams

- There are **4 Exams** in total. **Exams 1–3** each contribute **8%** toward the final course grade, and the **Final Exam** contributes **10%**, for a total of **34%** of the course grade.
- Exams will be administered on the scheduled class period.
- All exams are closed-book and closed-notes.
- **No make-up exam will be offered for the final exam.**
- **Make up Policy:** Students who miss a scheduled exam (Exam 1, 2 or 3) **do not** need to request a make-up exam. A single cumulative make-up exam will be offered during Exam Week. Students may take a single cumulative make-up exam during Exam Week, regardless of whether they missed or completed Exams 1, 2, or 3.
- **This make-up policy does not apply to the final exam; no make-up will be offered for the final exam.**
    - Each student has **one** make-up opportunity and may use the cumulative exam to replace the score of **one or two** of the first three exams.
    - The cumulative exam score will replace the original exam score(s) regardless of whether it is higher or lower than the original score (s).
    - No additional make-up exams will be given.

Projects

- There are **three (3)** projects in total.
- All projects are individual assignments unless otherwise stated.
- All project submissions must be completed through Canvas (LMS).
- Each project is worth 100 points for undergraduate students. Graduate students are required to complete an additional 25-point graduate component for each project (total possible score: 125 points). Graduate project scores will be prorated to a 100-point scale.
- Undergraduate students may complete the graduate component for extra credit, as specified in the project guidelines.
- **AI policy**: The use of AI tools is generally prohibited unless explicitly allowed in the project guidelines.
- **Make-up Policy:** One project make-up submission will be allowed during the last two weeks of the semester.
- **Late Policy**: Each project will have a Due Date and an Available Until date. The Available Until date serves as a no-questions-asked late/accommodation window for all students, including students with DRC extended-time accommodations. After the Available Until date, no submissions will be accepted.

AI Policy

Definition. Generative AI (Gen AI) refers to advanced machine learning models trained on vast datasets to predict and assemble data patterns into original outputs. It appears in code generators such as GitHub Copilot, OpenAI's Codex, and Claude Sonnet.

Appropriate Use. AI permissions will be specified for each assessment. Depending on the activity, AI may be used to brainstorm, learn/explain, debug, improve, or generate code. If an assessment states **No AI**, GenAI assistance should not be used.

Rationale. CS 411 is about understanding AI as well as learning to use it responsibly. AI should support your learning, not replace the learning that an assessment is designed to measure.

Disclosure. When AI is permitted and used, students must provide an **AI Use Declaration** identifying the tool used and briefly explaining how it was used.

Detection & Consequences. Students must be able to explain and defend all submitted work. Undisclosed or unauthorized AI use may be reviewed through submitted work, code history, interviews, or other available evidence and may be treated as an academic integrity violation.

Discussion. AI tools and acceptable practices continue to evolve. If you are uncertain whether a particular use of AI is permitted, ask the instructor or TA before using it. Questions and discussion about responsible AI use are encouraged.

Attendance / Participation Policy

Please email me timely if you face an unexpected situation that may impede your attendance, participation in required class, labs and exam sessions, or timely completion of assignments.

Academic Integrity

As a student and member of the UIC community, you are expected to adhere to the Community Standards of academic integrity, accountability, and respect. Please review the UIC Student Disciplinary Policy for additional information.

Email Expectations

Students are responsible **for all information instructors send to your UIC email and LMS** accounts. Faculty messages should be regularly monitored and read in a timely fashion.

IV. COURSE SCHEDULE

Weekly Schedule of Class Topics, Assignments, Assessments, Due Dates, and Deadlines

Week	Topics	Due/Exam
1	Introduction to Artificial Intelligence & Intelligent Agents	
2	Problem Solving by Search: Problem Formulation & Uninformed Search	
3	Uninformed Search (Continue) + Labor Day	
4	Informed Search (Heuristic Search, A*) • **Project 1 Opens**	Exam 1
5	Informed Search (Continue) & Adversarial Search	
6	Adversarial Search (Continue) & Constraint Satisfaction Problems	Project 1: Due
7	Constraint Satisfaction Problems • **Project 2 Opens**	
8	Constraint Satisfaction Problems: Continue	Exam 2
9	Bayesian Networks & Probabilistic Inference	Project 2: Due
10	HMM & Intro to Machine learning • **Project 3 opens**	

Week	Topics	Due/Exam
11	Naïve Bayes, Model Evaluation and Overfitting	Exam 3
12	Neural Networks & Deep Learning	Project 3: Due
13	Reinforcement Learning	
14	LLM, Generative AI & Agentic AI	
15	AI Ethics & Safety (Continue), Exam Review	

Refer to the course calendar page for updated and details.

Disclaimer

This syllabus is intended to give the student guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance through in-class announcements and in writing via LMS Announcements.

V. ACCOMMODATIONS

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations. Students are responsible for providing their LOA to the instructor in advance so that accommodations can be arranged.

Religious Accommodations

Following campus policy, if you wish to observe religious holidays, you must notify me by the **tenth day** of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least **five days** before you will be absent. Please submit this form by email with the subject heading: **"YOUR NAME: Requesting Religious Accommodation."**

Assignment Extended Time /Late Policy Accommodations

- **Quiz Extra Time**: In-class quizzes are normally due by **5:30 PM** and remain available until **6:30 PM on the same day**. Submissions after 5:30 PM are considered **late** unless the student has an approved quiz extended-time accommodation. Students with an approved LOA who have provided it to the instructor and informed the instructor in advance may use the extended availability period without a late penalty, consistent with their approved accommodation. No quiz submissions will be accepted after 6:30 PM.
- **Lab Extra Time:** In-class labs are normally due by 6:30 PM on the same day. Students who have submitted an approved LOA to the instructor will receive an extended submission deadline until 11:59 PM on the same day, without a late penalty. This extended deadline is provided only to students with an approved LOA and is not available by default. No lab submissions will be accepted after 11:59 PM.
- **Exam Extra Time and Testing Accommodations**: Exams are designed for approximately one hour of testing time. Students requiring extended testing time and/or other testing accommodations (e.g., a reduced-distraction or separate testing environment) must provide their LOA and contact the instructor **at least five business days in advance**. Testing arrangements will be determined based on the accommodations specified in the LOA and, when necessary, coordinated with the DRC.
- **Project Late Policy:** Each project will have a **Due** Date and an Available **Until** date. The Available Until date serves as a no-questions-asked late/accommodation window for all students, including students with DRC extended-time accommodations. After the Available Until date, no submissions will be accepted. For documented extraordinary circumstances (e.g., hospitalization for multiple days), students must contact the instructor by email to request consideration for additional accommodations.

VI. CLASSROOM ENVIRONMENT

Inclusive Community

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

Name and Pronoun Use

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are *[she/her; he/him; they/them]*. I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: https://www.mypronouns.org/what-and-why.

Community Agreement/Classroom Conduct Policy

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another's privacy, and do not invite outsiders into our classroom.

Content Notices and Trigger Warnings

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

VII. RESOURCES: Academic Success, Wellness, and Safety

We all need the help and the support of our UIC community. Please visit my **office hours** for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

Academic Success

- UIC Tutoring Resources
- College of Engineering tutoring program
- Equity and Inclusion in Engineering Program
- UIC Library and UIC Library Research Guides.
- Offices supporting the UIC Undergraduate Experience and Academic Programs.
- Student Guide for Information Technology

Wellness

- **Counseling Services**: You may seek free and confidential services from the Counseling Center at https://counseling.uic.edu/.
- Access U&I Care Program for assistance with personal hardships.
- **Campus Advocacy Network**: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at http://can.uic.edu/.

Safety

- UIC Safe App—PLEASE DOWNLOAD FOR YOUR SAFETY!
- UIC Safety Tips and Resources
- Night Ride
- Emergency Communications: By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.